Programming Project: Task Scheduler

Instructions:

- **The project requires completing tasks by May 08, 2025.**
- **Your goal is to implement a Task Scheduler as a prompt window that allows users to manage a list of tasks. The task list must be internally stored using a linked list(s).**
- **A task display in a list must be as follows**

$$\begin{cases} \texttt{"}id\texttt{. [ ] }title\texttt{ (}priority\texttt{)"} & \text{if task is incompleted} \\ \texttt{"}id\texttt{. [X] }title\texttt{ (}priority\texttt{)"} & \text{if task is completed} \end{cases}$$

- **An individual task display must be as follows**

$$\begin{cases} \texttt{"}id\texttt{. [ ] }title\texttt{ (}priority\texttt{)\textbackslash n}description\texttt{"} & \text{if task is incompleted} \\ \texttt{"}id\texttt{. [X] }title\texttt{ (}priority\texttt{)\textbackslash n}description\texttt{"} & \text{if task is completed} \end{cases}$$

- **The scheduler must support and have a prompt command for the following operations:**
  - **Add a task (at the beginning, end, or a specific position).**
  - **Remove a task (by ID or position).**
  - **Edit a task's content (not including ID).**
  - **Mark a task as complete/incomplete.**
  - **Reorder tasks (move up/down in the list).**
  - **Display all tasks.**
  - **Display a detailed individual task.**
  - **Display a list of prompt commands.**
- **You must use the *Node* class from 'Node.h'.**
- **Each task must have at least the following attributes:**
  - **Unique ID generated when the task is created**
  - **Title**
  - **Description**
  - **Priority level that is either Low, Medium, or High.**
  - **Completion status (complete/incomplete)**
- **All classes that display content must inherit *Object*.**
- **All classes must be defined in header files.**
- **The header files can only use the libraries *iostream*, *string*, *sstream*, *cctype*, *iomanip*, *cmath*, 'Object.h', 'Node.h'. and any header file your define.**
- **Your final submission should include:**
  - **the source codes with comments.**
  - **documentation file (.pdf or .md) that contains:**
    - **a list and description of all classes you implemented such that pseudocode is provided for each method.**
    - **a brief explanation of the overall program design.**
    - **instructions on how to run your program.**
- **Cheating of any kind is prohibited and will not be tolerated.**
- **Violating and/or failing to follow any rules will result in an automatic zero (0) for the project.**

Hints & Suggestions

- You should implement the classes
  - Task – Represents a single task item.
  - TaskList – Custom linked list class that handles adding, removing, editing, and moving tasks.
  - TaskScheduler – Main class provides the prompt window and interacts with the task list.